Project 6 – Indicator Evaluation

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1 INDICATORS

The following chosen technical indicators and their associated graphs represent JPM stock data in the date range of 2008-01-01 to 2009-12-31.

Exponential Moving Average

The Exponential Moving Average (EMA) is a moving average that places higher weight on recent data points, making it more responsive to price changes compared to the Simple Moving Average (SMA). Calculating the EMA involves an initial SMA value for the first period, followed by subsequent values using a formula that smoothens the price series by weighting recent prices more heavily. For JPM stock, the EMA helps identify trends and potential buy or sell signals; a price crossing above the EMA may indicate a bullish trend, while a price crossing below may indicate a bearish trend. The EMA can be used alone or with other indicators, such as in a crossover strategy with two EMAs of different periods, to generate reliable trading signals. This sensitivity to recent price movements makes the EMA a valuable tool for analyzing stock price trends and making informed decisions in trading stocks. The formula is as follows:

$$\mathrm{EMA}_t = \mathrm{Price}_t \times \frac{2}{n+1} + \mathrm{EMA}_{t-1} \times \left(1 - \frac{2}{n+1}\right)$$

Where EMA_t is the EMA at time t and Price_t is the price at time t, and n being the number of periods.

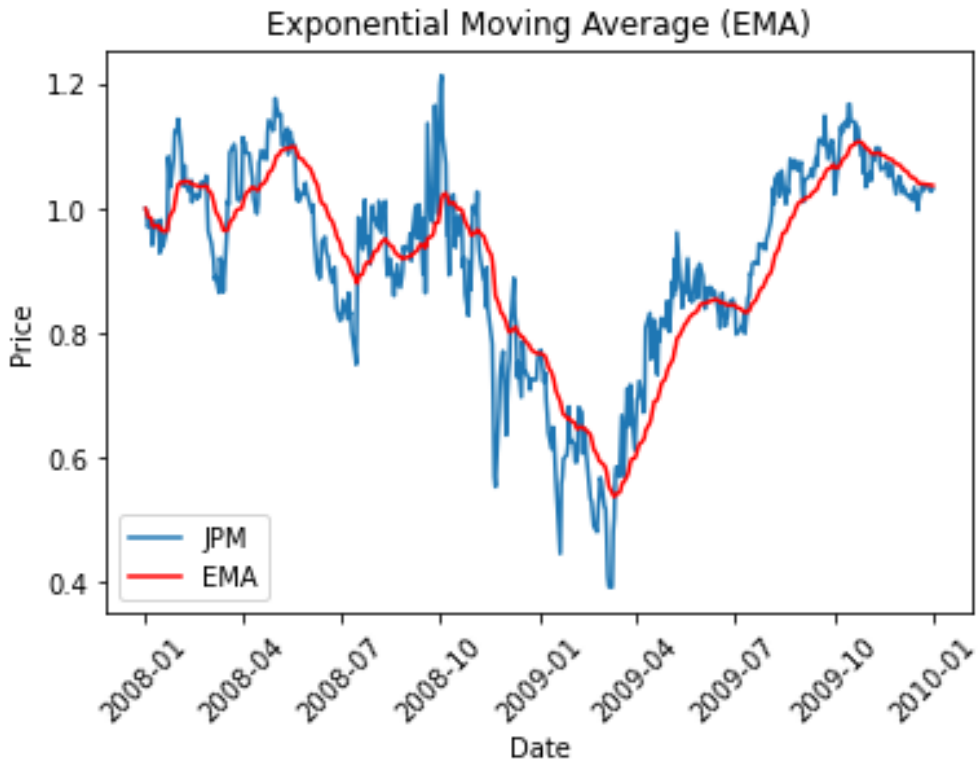

Figure 1. JPM stock price comparison to Exponential Moving Average of a 30 day window.

Relative Strength Index

The Relative Strength Index (RSI) is a momentum oscillator used to measure the speed and change of price movements in a specific stock. It ranges from 0 to 100, with readings above 70 generally considered overbought and readings below 30 considered oversold. The RSI is calculated by comparing the magnitude of recent gains to recent losses over a specified period, typically 14 days, using the formula:

$$RSI = 100 - \left(\frac{100}{1+RS}\right)$$

where RS (Relative Strength) is the average of X days' up closes divided by the average of X days' down closes.

For the JPM stock price, the RSI can provide valuable insights into potential price reversals or continuations. An RSI above 70 may suggest that the stock has been overbought and could be due for a pullback, making it a signal to consider selling or shorting the stock. Conversely, an RSI below 30 may indicate that the stock is oversold and could be poised for a rebound, serving as a potential buy signal. The RSI can be particularly effective when used in conjunction with other indicators, such as moving averages or MACD, to confirm signals and enhance trading strategies. By identifying overbought and oversold conditions, traders can make more informed decisions about entry and exit points, potentially improving their overall trading performance.

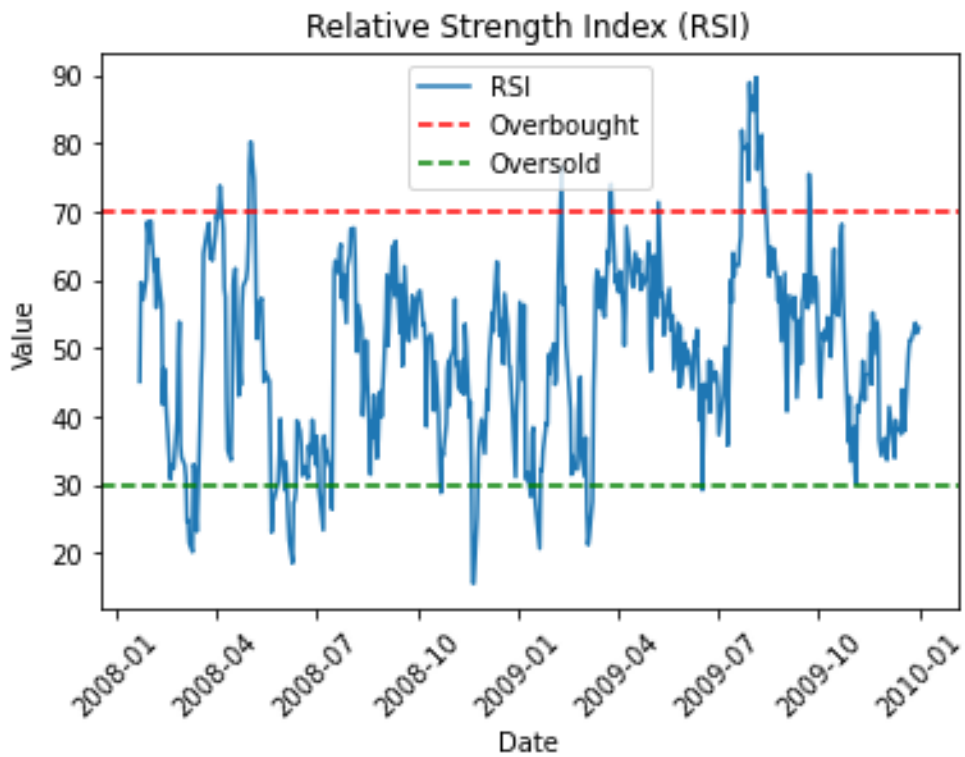

Figure 2. JPM RSI values with a window span of 14 days.

Bollinger Bands

Bollinger Bands are a technical indicator used to measure market volatility and identify potential buy and sell signals. They consist of a simple moving average (SMA) and two standard deviation lines, one above and one below the SMA. To reproduce Bollinger Bands: calculate the SMA (typically 20 days), compute the standard deviation for the same period, and then calculate the upper band (SMA

+ 2 * SD) and the lower band (SMA – 2 * SD). This indicator is effective because it adjusts to market conditions, expanding during high volatility and contracting during low volatility. When the stock price touches the lower band, it suggests the stock may be oversold, indicating a potential buy signal, while touching the upper band suggests overbought conditions and a potential sell signal. Bollinger Bands can be used alone or in conjunction with other indicators like the Relative Strength Index (RSI) for more robust trading signals.

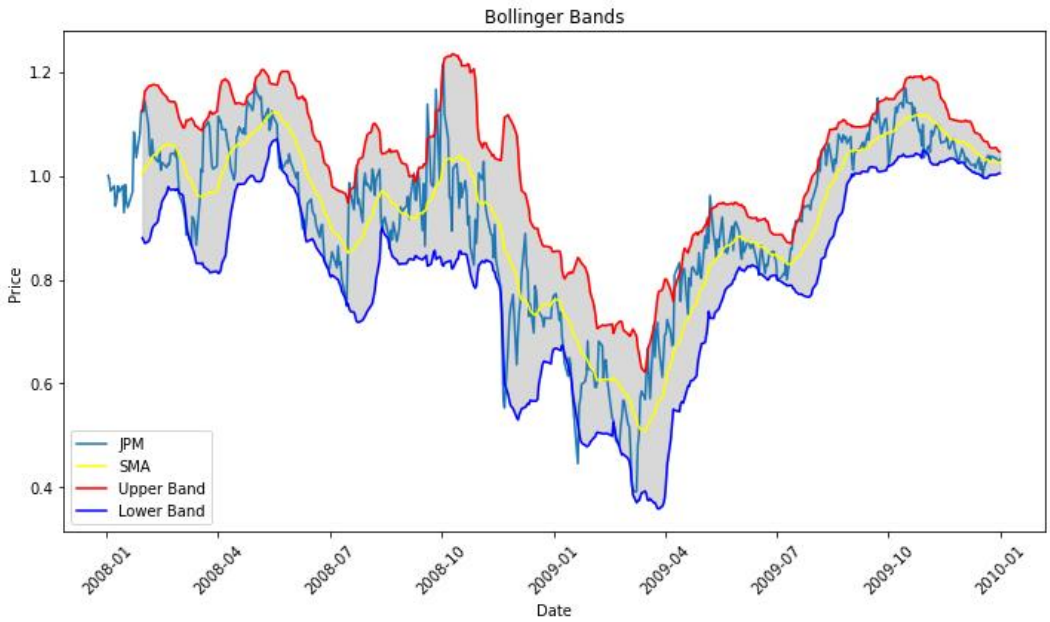

Figure 3. JPM Bolinger Bands with a rolling window of 20 days along JPM price.

Commodity Channel Index

The Commodity Channel Index (CCI) is a momentum-based technical indicator used to identify cyclical trends in a stock's price movement. To calculate CCI, first calculate the typical price (TP) as the average of the high, low, and close prices for a period (typically 20 days). Then, calculate the SMA of the typical price, and the mean deviation of the typical price from this SMA. Finally, CCI is derived using the formula: CCI = (TP – SMA) / (0.015 * Mean Deviation). CCI values above +100 suggest overbought conditions, indicating a potential sell signal, while values below -100 indicate oversold conditions, suggesting a potential buy signal. This indicator is useful as it helps traders identify overbought and

oversold levels, and it can be used in conjunction with other indicators like Moving Averages to enhance trading decisions.

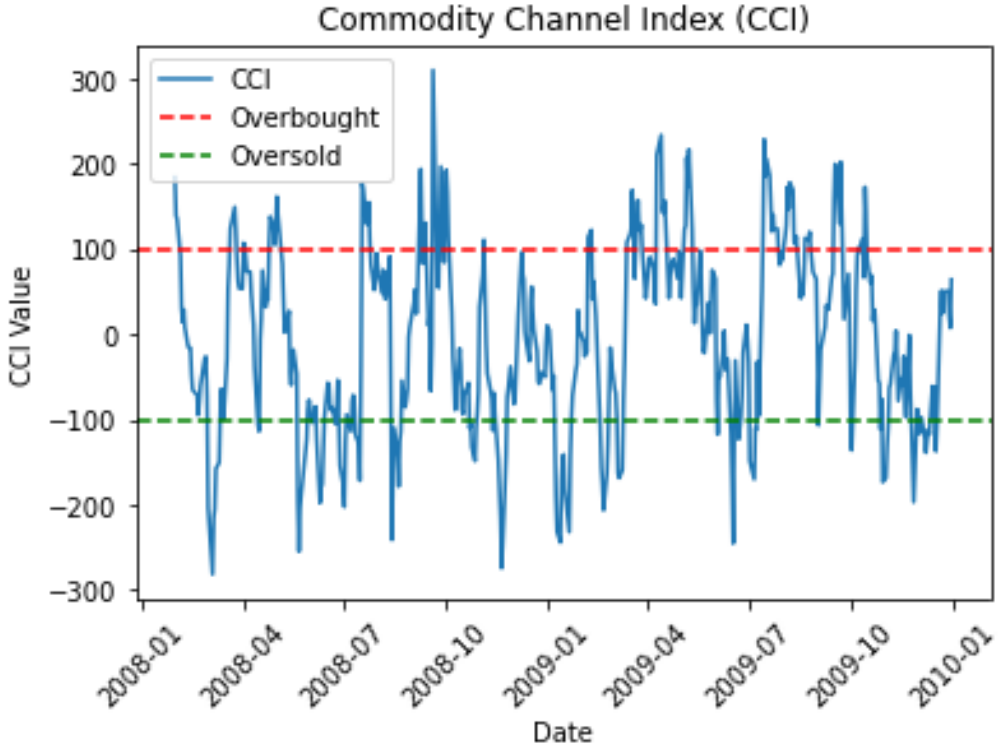

Figure 4. Commodity Channel Index values for JPM with a 20 day window span.

Moving Average Convergence Divergence

The Moving Average Convergence Divergence (MACD) is a trend-following momentum indicator that shows the relationship between two moving averages of a stock's price. It is calculated by subtracting the 26-period Exponential Moving Average (EMA) from the 12-period EMA. This difference is the MACD line. A 9-period EMA of the MACD line, called the signal line, is then plotted on top of the MACD line, which can function as a trigger for buy and sell signals. When the MACD line crosses above the signal line, it generates a buy signal, indicating upward momentum. Conversely, when the MACD line crosses below the signal line, it generates a sell signal, indicating downward momentum. The MACD is effective in identifying changes in the strength, direction, momentum, and duration of a trend, and it is often used alongside other indicators like RSI to confirm signals.

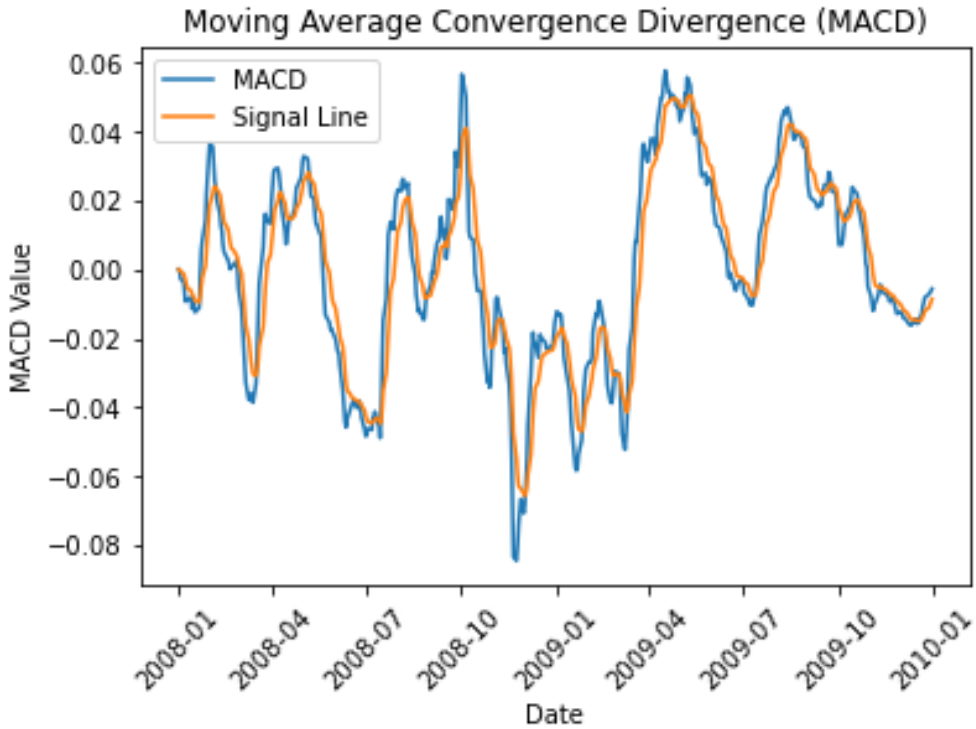

Figure 5. Moving Average Convergence Divergence value versus date indicating the signal line in yellow for JPM stock.

2 THEORETICALLY OPTIMAL STRATEGY (TOS)

For implementing the Theoretically Optimal Strategy (TOS), the goal is to maximize performance assuming perfect foresight within the constraints of trading 1000 shares long, 1000 shares short, or no position at all. Using a constructed market simulator with zero commissions and impact, trades are executed based on anticipated future price movements of JPM stock from January 1, 2008, to December 31, 2009. The resulting trades, housed in the Python `testPolicy()` function, generate a series of buy and sell signals aiming to achieve the best possible portfolio value starting with $100,000. This approach provides an upper bound on performance, contrasted against a benchmark holding 1000 shares of JPM throughout the period. Visualized on a chart, the benchmark and theoretically optimal portfolio are normalized to 1.0.

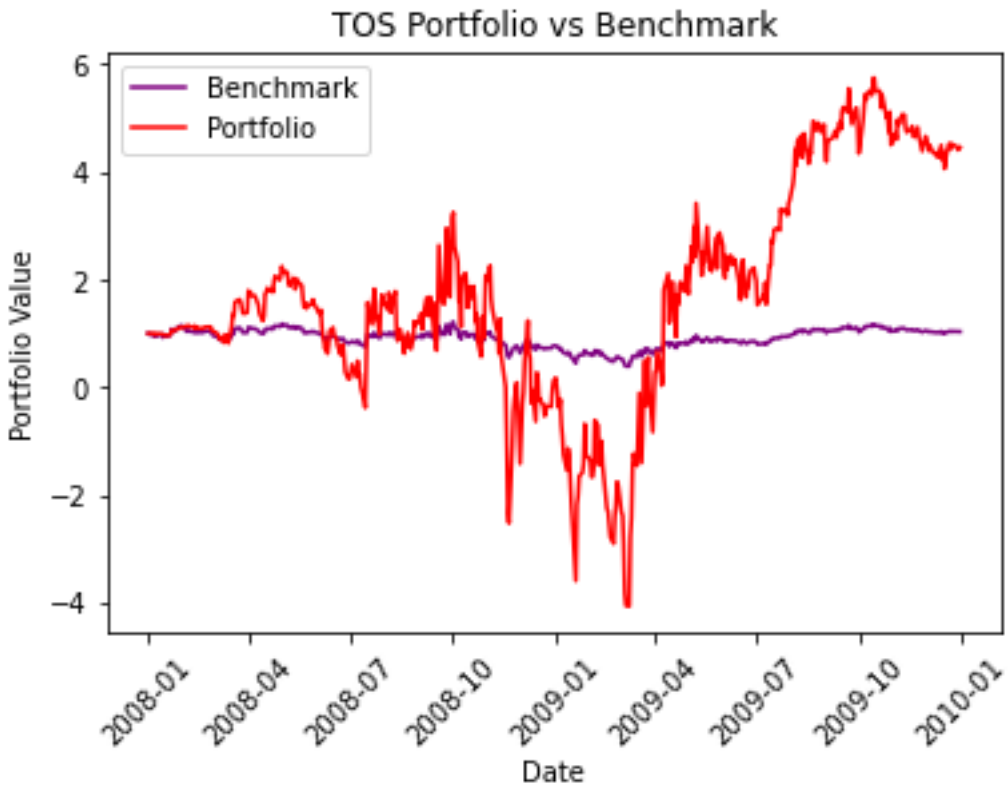

Figure 6. Mean Absolute Logarithmic Error(MALE) and Median Absolute Percentage Error(MdAPE) for DTLearner and RTLearner versus Leaf Size.

The TOS Portfolio has a great deal of volatility compared to Benchmark however outperforms Benchmark on total returns by end of period.

Table 1. Statistics for TOS Portfolio and Benchmark

Benchmark STD: 0.0417

Benchmark Cumulative Rets: 0.032

Benchmark Avg Returns = 0.0001

TOS STD: 0.392

TOS Cumulative Rets 3.4555

TOS Average Rets: 0.0069

3 REFERENCES

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[5] Dolan, B. (2024, March 8). *What is MACD?* Investopedia. https://www.investopedia.com/terms/m/macd.asp#:~:text=Key%20Takeaways-,Moving%20average%20convergence%2Fdivergence%20(MACD)%20is%20a%20technical%20indicator,EMA%20of%20the%20MACD%20line.